Khue (Cherry) Pham

Boston, MA | linkedin.com/in/cherryyypham | github.com/cherryyypham | khuee.phamj@gmail.com

Education

Northeastern University, The Roux Institute – M.S. Computer Science Dec 2027

Relevant Coursework: Programming Design Paradigms; Machine Learning; Database Management Systems

Olin College of Engineering – B.S. Electrical and Computer Engineering May 2025

Technical Skills

Languages: Python, SQL

Software Engineering: ETL data pipelines, FastAPI, Git

ML/AI: PyTorch, scikit-learn, Fine-tuning, Hugging Face Transformers, LLMs, NLP

Experience

Florida Center for Reading Research Remote

Machine Learning Engineer Sep 2024 – Aug 2025

- Built a PyTorch/LangChain pipeline and fine-tuned a LLaMA-based LLM on early literacy corpora, evaluating prompting and fine-tuning variants against rubric-aligned benchmarks to select version for cross-state pilot deployment.
- Built RAG pipelines with pgvector-backed semantic search (PostgreSQL), improving retrieval accuracy by 15–25% on children's reading development tasks; evaluated using BLEU/ROUGE benchmarks.
- Designed a hybrid SQL/NoSQL data model from an ERD, using relational storage for interconnected student, teacher, parent, classroom, and game data and document storage for variable per-game metrics; deployed on Red Hat OpenShift with OAuth 2.0 and role-based access control.

Air Partners Needham, MA

Data Engineer Jun 2023 – Aug 2023

- Designed a normalized data model and modular ETL pipeline to integrate time-series data from 18 disparate IoT sensors into an analysis-ready dataset, processing ≈3 million records across 2 years of hourly readings; reduced pipeline latency by 30% through parallelized processing and optimized API batching.
- Consolidated scattered Python and R scripts into a reproducible R workflow with documentation, enabling non-technical team members to modify and rerun the pipeline independently.

Olin College – College as a Living Lab Needham, MA

AI Pedagogy Researcher Jun 2025 – Aug 2025

- Evaluated prompting strategies across 4–6 model iterations against professor-graded rubrics, reaching 80–87% alignment with human graders through iterative prompt design and layer-specific fine-tuning.
- Designed and deployed a GPT-4o-based automated grading pipeline integrated with the Canvas LMS API, processing 500–1,000 student submissions per semester with tokenization and semantic similarity analysis.

Projects

ENCoM – Real-Time Generative Speech-to-Speech System | GitHub

- Designed and ran experiments comparing a Mimi+Mamba hybrid architecture against a controlled cascaded baseline, achieving 2.79x inference speedup while maintaining audio quality (MOS) on Switchboard/Expresso evaluation sets.
- Conducted ablation studies on architecture variants, measuring perplexity for same-condition comparisons and automated speech-quality scoring via Whisper transcription for cross-condition evaluation.
- Benchmarked end-to-end performance against SOTA cascaded pipelines (Whisper/GPT/Kokoro), delivering a 3x improvement in RTFx and significantly higher barge-in recovery precision.

Next-Step – Full-Stack School Management Platform | GitHub

- Designed a cross-platform REST API with role-based access control for 3 user roles using async Python (FastAPI, SQLAlchemy 2.0).
- Containerized with Docker and deployed via multi-worker uvicorn, validated at 100 concurrent users with 7,801 requests served, zero errors, and 95% of responses under 200ms.